

Accounting for Love of Variety in Consumer Demand Estimation

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1 Introduction

In the baseline logit model, a consumer is perceived as having a fixed preference for each product, and the utility from consuming a product is independent of past consumption. Further, it is assumed that a new good is always a utility boosting event, by the error term's property of being identically and independently distributed. However, in many markets, consumers may have a love of variety, or a desire to consume different products over time. This can be due to factors such as boredom, satiation, or a preference for novelty. In such cases, the utility from consuming a product may depend on past consumption, and the introduction of a new good may not always lead to an increase in utility.

In this paper, I build upon the existing love of variety literature by modernizing the maximum likelihood methods as well as using moment estimation to get aggregate estimates of demand. I also plan to see how these preferences for variety differ across product spaces. Using Nielsen data, I estimate a discrete choice demand model using Nielsen scanner data. Consumers are myopic, only internalizing past consumption as an input for variety without strategically choosing products.

I find that...

The rest of the paper is organized as follows: section two discusses the related literature and section three describes the data used and some descriptive statistics. Section four showcases the model and environment and section five presents results. Section six goes into counterfactual analysis, while section 7 concludes.

2 Related Literature

The variety-seeking and state dependence literature has been primarily focused on understanding the dynamics of consumer preferences and demand over time. From ?, who developed an empirical structural model of state dependence in consumer choice, to ?, who iterated the model to allow for consumer learning. ? is the most closely related paper, as he develops a model of variety-seeking behavior which is both tractable and flexible, and uses Nielsen data to estimate the model. However, this paper only looks at demand, leaving me a niche to explore the interplay between variety-seeking and strategic advertising.

3 Data

3.1 Data

My data comes from Nielsen's HomeScan and MarketScan data. These two main datasets give me access to household purchase decisions as well as product characteristics. I am looking at the years between 2014-2016 in markets Atlanta, Chicago, San Diego, and Denver. Households are single-person to avoid any contamination of shopping for multiple people like children or spouses, allowing me a cleaner sample of shopping trips.

The product space I am looking at is yogurt. Yogurt is good for this sort of analysis due to its relative lack of storability as well as its variation in attributes like flavor, nutrient contents, and types like Greek or Skyr. I have limited yogurt sizes to two kinds: cups and tubs. Cups of yogurt are between five and seven ounces while tubs are approximately 32 ounces. Cups are single serving products and come in more varied flavors. Tub, on the other hand, tend to be limited in flavors but hold five to seven servings of yogurt. These two types provide interesting analysis of what type of person buys which size.

?? shows that there are n households, with a in Atlanta, b in Chicago, c in Denver, and d in San Diego. The mean income is \$17,000 and the mean age is n years old. The sample is mostly white at $n\%$. Households go on an average of n shopping trips, indexed by a unique trip code. Per trip, they buy an average of n units of yogurt. If we break out by type, n cups and/or n tubs.

I define the outside option to be anytime yogurt isn't purchased during a shopping occasion within a trip. In each trip, the outside option was taken $x\%$, on average.

4 Model

4.1 Environment

There are a continuum of $i \in \mathcal{I}$ consumers who exist for $\mathcal{T}$ periods. Within a given period, a household can go for a shopping trip $t \in \mathcal{T}$ which is comprised of choice occasions $o \in \mathcal{O}_t$. Each choice occasion allows for the purchase of a singular good.

4.2 Consumer's Problem

The consumer's problem is the following. In each period t , the consumer i chooses a product j from the menu of available products J_t . Each product $j \in J_t$ is a bundle of $k \in K$ characteristics such that $X_{jt}^k \in \mathcal{X}_{jt}$. The consumer's utility from choosing product j at time t is given by:

$$U_{ijt} = \delta D_{ht} + \sum_1^k \beta_i^k X_{jt}^k + \gamma_i \log(1 + \Xi_{ijt}) + \alpha_i p_{jt} + \xi_j + \varepsilon_{ijt}$$

subject to

$$\begin{aligned}\Xi_{ijt} &= \sqrt{\sum_k (X_{jt}^k - \theta_{ijt}^k)^2} \\ \theta_{ijt}^k &= \lambda \frac{1}{t-2} \sum_s^{t-2} X_{js}^{*k} + (1-\lambda) X_{jt-1}^{*k} \\ \gamma_i &\sim \mathcal{N}(0, 1) \\ \varepsilon_{ijt} &\sim \text{T1EV}(0, 1)\end{aligned}$$

where δD_{ht} are household specific effects, β_i^k is a consumer's utility from characteristic X_{jt}^k , γ_i is their preference for variety, Ξ_{ijt} is the difference between the current characteristic and the previous choices, θ_{ijt}^k . θ_{ijt}^k is a function of the consumers' past purchases, where λ weights chosen products in the past, X_{js}^{*k} and the most recently chosen product, X_{jt-1}^{*k} , allowing for more recent purchases to play more of a role in the current period's decision. This could take many forms, but in this paper will be the mean characteristic of past purchases. α_i is a heterogeneous price sensitivity to p_{jt} . ξ_j is unobserved quality in product j , and ε_{ijt} is the Gumbel shock which allows for a closed form logit.

The choice probability of consumer i choosing product j in time t is as follows:

$$P_{ijt} = \frac{\exp U_{ijt}}{\sum_k \exp U_{ikt}}$$

5 Results

6 Counterfactuals

6.1 Counterfactual 1

6.2 Counterfactual 2

Secondly, I am interested in pivoting to a more consumer welfare focused counterfactual, in which I evaluate the effects of product entry on consumer welfare under different levels of γ . This will allow me to understand how a love of variety affects consumer welfare in the face of new product entry, and to evaluate the potential benefits and drawbacks of a love of variety for consumers in dynamic markets. Specifically, I want to understand how consumer welfare is impacted by both perfect and obscured information about the distribution of γ when introducing a new product. I expect to find that a love of variety can lead to higher consumer welfare in the face of new product entry, as it allows consumers to derive more utility from a wider range of products. However, if consumers have obscured information about the distribution of γ , they may not be able to fully realize the benefits

of new product entry, which could lead to lower consumer welfare.

7 Conclusion

In closing, this paper develops a dynamic discrete choice model of consumer demand that incorporates a love of variety and strategic advertising by firms. The model is estimated using scanner data, and counterfactual analyses are conducted to evaluate the effects of different market structures on consumer welfare and market outcomes. The paper contributes to the literature on advertising and consumer behavior, by providing empirical evidence on the effects of strategic advertising in a dynamic demand setting. The findings of this paper have important implications for policymakers and firms, as they highlight the potential benefits and drawbacks of targeted advertising in markets when accounting for consumers' preference for variety.